Chapter 2

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1 2-1

We can describe objects in an idealized manner in the representation of a 'particle' To describe the motion of a particle, we need to describe the 'position' of the particle and how position changes as the particle moves. The change in something can be called displacement, so $x_f - x_i = \Delta x$. Average speed is equal to $\frac{totaldistance}{totaltime} = \frac{s}{\Delta t}$. Velocity is speed and direction and the 'average velocity' is represented as $v_{av\,x} = \frac{\Delta x}{\Delta t} = \frac{x_f - x_i}{t_f - t_i}$ so $\Delta x = v_{av\,x}\Delta t$. The next part is about slope but like everyone knows what slope is.

The instantaneous velocity v_x is the limit of the ratio $\Delta x/\Delta t$ as Δt approaches 0. $v_x(t) = \lim_{\Delta t \to 0} \frac{\Delta x}{\Delta t}$

This is literally just a derivative of dx/dt. This is instantaneous velocity.

2 2-2

Acceleration is the rate of change of velocity with respect to time. The average acceleration, $a_{av\,x}$ for a particular interval Δt is defined as the change in velocity Δv divided by that time interval. $a_{av\,x} = \frac{\Delta v_x}{\Delta t} = \frac{v_{fx} - v_{ix}}{t_f - t_i}$ Instanenous acceleration is the limit of the ratio Δv_x / Δt as Δt approaches 0. Acceleration therefore is the derivative of velocity v_x with respect to time dv_x/dt. Velocity is the derivative of the position x with respect to t, acceleration is the second derivative of x with respect to t, d^2x/dt^2. When v_x and a_x are both positive and becoming more positive the speed is increasing. If they are both negative, and v_x is becoming more negative the speed is again increasing. When they have oppposite signs the object is slowing down. If velocity is positive and acceleartion is negative and velocity is positive but becoming less positive the speed is decreasing. If velocity is negative and acceleration is positive and velocity is becoming less negative the speed is again decreasing. Basically, if they have the same sign the speed is increasing, if opposite the speed is decreasing.

3 2-3

For a moving particle the final velocity v_x equals the inital velocity plus the change in velocity and the change in velocity equals the average acceleration multiplied by time.

$v_x = v_{0x} + \Delta v = v_{0x} + a_{av\,x}\Delta t$

If a particle has constant acceleration a_x it follows that the instantaneous acceleration and the average acceleration are equal.

$a_x = a_{av\,x}$ (a_x is constant)

These equations involving nearly constant acceleration are common and so we can use these to derive a set of **kinematic equations** for problems involving one-dimensional motion at constant acceleration.

If a particle is moving with constant acceleration a_x has velocity v_{0x} at time $t_0 = 0$ and v_x at some time later t. $v_x = v_{0x} + a_x t$

And, $\Delta x = v_{1x}\Delta t + \frac{1}{2}a_x\Delta t^2$

If we set $t_1 = 0$ and $t_2 = t$ then the equation above becomes $x - x_0 = v_{0x}t + \frac{1}{2}a_x t^2$

4 2-4

As we know, $dv_x(t)/dt = a_x(t)$ meaning that the derivative of velocity with respect to time is equal to acceleration. If acceleration is constant, the velocity of that function of time when differentiated equals the constant. $v_x = a_x t$ a_x is constant $v_x = a_x t + c$ When t = 0, $v_x = c$. Therefore c is the velocity v_{0x} at time $t = 0$. The position function $x(t)$ is that function whose derivative is the velocity: $dx/dt = v_x = v_{0x} + a_x t$.

You can treat each term separatly, $x = x_0 + v_{0x}t + \frac{1}{2}a_x t^2$. In that equation, when $t = 0, x = x_0$. Usually we are given the ***initial conditions*** of a situation which is called the initial value problem, and takes form like given acceleration and the init values find x(t).

The sum of the displacement during a set of time intervals is approximately equal to the total displacement from time t_1 to time t_2.

$$\Delta x \approx \sum_i v_{ix}\Delta t_i$$

An alternative definition of average velocity is $v_{av\,x} = \frac{1}{\Delta t}\int_{t_1}^{t_2} v_x dt$

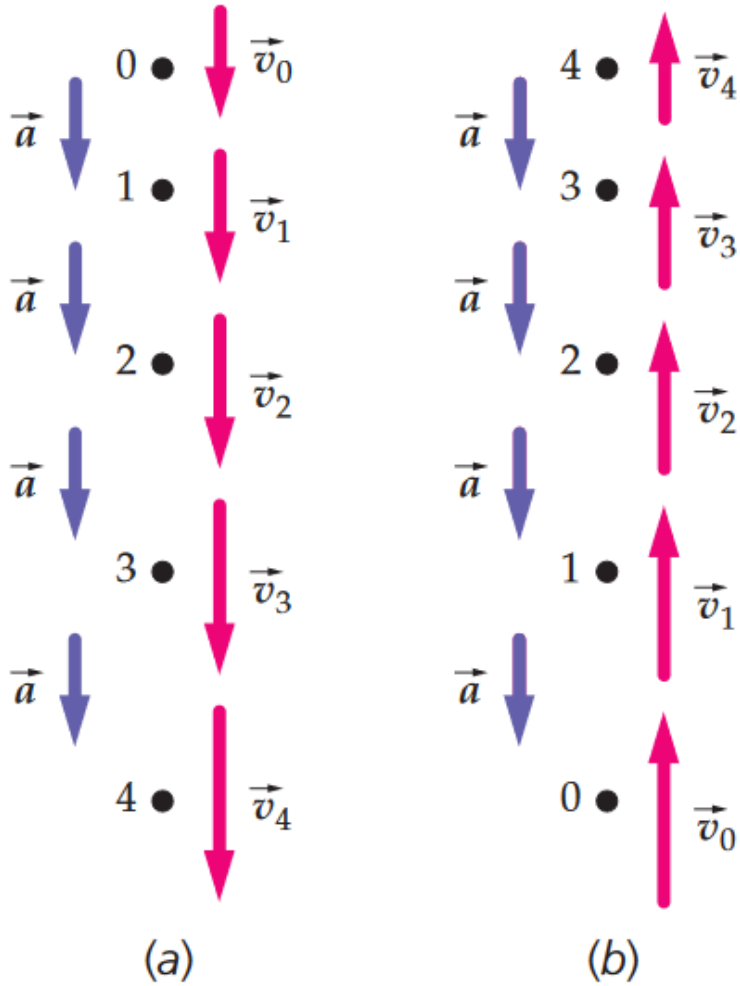

FIGURE 2-9 Motion diagrams. The time intervals between successive dots are identical. (*a*) The velocity vector is increasing, so the acceleration is in the direction of the velocity vector. (*b*) The velocity vector is decreasing, so the acceleration is in the direction opposite to that of the velocity vector.

Figure 1: Motion Diagrams

The antiderivative of a function is also called the indefinite integral of the function and is written without limits on the integral sign, as in

$$x = \int v_x \, dt$$

Finding the function x from its derivative v_x (that is, finding the antiderivative) is also called integration. For example, if $v_x = v_{0x}$, a constant, then

$$x = \int v_{0x} \, dt = v_{0x}t + x_0$$

where x_0 is the arbitrary constant of integration. We can find a general rule for the integration of a power of t from Equation 2-6, which gives the general rule for the derivative of a power. The result is

$$\int t^n \, dt = \frac{t^{n+1}}{n+1} + C, \quad n \neq -1 \qquad \text{2-19}$$

where C is an arbitrary constant. This equation can be checked by differentiating the right side using the rule of Equation 2-6. (For the special case $n = -1$, $\int t^{-1} dt = \ln t + C$, where $\ln t$ is the natural logarithm of t.)

Because $a_x = dv_x/dt$, the change in velocity for some time interval can similarly be interpreted as the area under the a_x-versus-t curve for that interval. This change is written

$$\Delta v_x = \lim_{\Delta t \to 0} \left(\sum_i a_{ix} \Delta t_i \right) = \int_{t_1}^{t_2} a_x \, dt \qquad \text{2-20}$$

We can now derive the constant-acceleration equations by computing the indefinite integrals of the acceleration and velocity. If a_x is constant, we have

$$v_x = \int a_x dt = a_x \int dt = v_{0x} + a_x t \qquad \text{2-21}$$

where we have expressed the product of a_x and the constant of integration as v_{0x}. Integrating again, and writing x_0 for the constant of integration, gives

$$x = \int (v_{0x} + a_x t) dt = x_0 + v_{0x}t + \tfrac{1}{2}a_x t^2 \qquad \text{2-22}$$

It is instructive to derive Equations 2-21 and 2-22 using definite integrals instead of indefinite ones. For constant acceleration, Equation 2-20, with $t_1 = 0$, gives

$$v_x(t_2) - v_x(0) = a_x \int_0^{t_2} dt = a_x(t_2 - 0)$$

where the time t_2 is arbitrary. Because it is arbitrary, we can set $t_2 = t$ to obtain

$$v_x = v_{0x} + a_x t$$

where $v_x = v_x(t)$ and $v_{0x} = v_x(0)$. To derive Equation 2-22, we substitute $v_{0x} + a_x t$ for v_x in Equation 2-17 with $t_1 = 0$. This gives

$$x(t_2) - x(0) = \int_0^{t_2} (v_{0x} + a_x t) dt$$

This integral is equal to the area under the v_x-versus-t curve (Figure 2-26). Evaluating the integral and solving for x gives

$$x(t_2) - x(0) = \int_0^{t_2} (v_{0x} + a_x t) dt = v_{0x}t + \tfrac{1}{2}a_x t^2 \Big|_0^{t_2} = v_{0x}t_2 + \tfrac{1}{2}a_x t_2^2$$

where t_2 is arbitrary. Setting $t_2 = t$, we obtain

$$x = x_0 + v_{0x}t + \tfrac{1}{2}a_x t^2$$

where $x = x(t)$ and $x_0 = x(0)$.

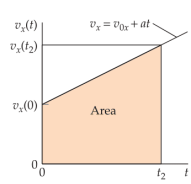

FIGURE 2-26 The area under the v_x-versus-t curve equals the displacement $\Delta x = x(t_2) - x(0)$.

Figure 2: Laziness